

India - France Adopt Innovation Roadmap 2030



For Prelims: Artificial Intelligence Impact Summit 2026 , Critical Technologies , Small Modular Reactors (SMRs) , SHANTI Act, 2025 , Shakti (Army) , Varuna (Navy) , Garuda (Air Force) , FDI , General Data Protection Regulation (GDPR) , Critical Minerals .

For Mains: Key highlights of the India-France Strategic Partnership, key areas of concern in the India-France relations and way forward.

Source: TH

Why in News?

The Indian Prime Minister and French President **Emmanuel Macron** held high-level bilateral talks in **Nice** , marking their first official meeting since the elevation of India-France ties to a **Special Global Strategic Partnership** .

- During the visit, the two leaders adopted the **Innovation Roadmap 2030** , and launched a first-of-its-kind Economic Security Dialogue.

Summary

- **India and France** adopted the **Innovation Roadmap 2030** and launched an **Economic Security Dialogue** to strengthen cooperation in AI, semiconductors, critical minerals, defence, nuclear energy, startups, education, and Indo-Pacific security.
- **Despite strong strategic convergence, challenges remain** due to delays in the Jaitapur Nuclear Power Project, differing digital regulations, geostrategic priorities, and low trade volumes; both countries aim to address these through technology co-development, resilient supply chains, maritime cooperation, and deeper economic engagement.



What are the Key Highlights of the Visit?

- **Adoption of the Innovation Roadmap 2030:** This is a comprehensive operational blueprint designed to secure the technological sovereignty of both nations, shifting the focus towards disruptive innovations and digital infrastructure.
 - **'Trusted AI' Alliance:** Setting up a **Joint India-France AI Working Group** to focus on safe, ethical, and risk-based **AI governance**.
 - **Globalisation of UPI:** Expanding the footprint of **India's Unified Payments Interface (UPI)** across major French hubs, including the Paris airport and Nice.
 - **Kanpur Aeronautics Center:** Establishment of a **National Centre of Excellence for Skilling in Aeronautics** at the National Skill Training Institute in Kanpur to bolster aerospace manufacturing readiness.
 - **Station F & Startup Linkages:** France will incubate 10 additional Indian startups at Station F in Paris (one of Europe's largest startup campuses) and aims to host 30,000 Indian students by 2030.
- **Launch of the Economic Security Dialogue:** The dialogue covers supply chain resilience in strategic sectors, including **critical minerals, semiconductors, energy, and cybersecurity**.
 - A high-level annual mechanism was established with the goal of **doubling bilateral trade** between India and France within the next five years.
- **Bharat Innovates 2026:** Organized by the **Union Ministry of Education**, it is a three-day event in Nice showcasing **120 Indian deep-tech startups across 13 critical technology pillars to global investors**.
 - The Indian Prime Minister emphasized that the technological revolution must be guided by **inclusive and human-centric technology**, noting that startups should be judged by their impact on humanity, not just market valuation.
- **Enhancing Student and Professional Mobility:** French universities have been invited to set up physical campuses in India under the **National Education Policy (NEP) 2020 framework**.

What are the Key Pillars of the India-France Strategic Partnership?

- **Historical Foundations:** Diplomatic relations were established in 1947, and the **Strategic Partnership was launched in 1998**, making it India's first-ever strategic partnership with a Western

nation.

- The partnership is anchored on three traditional pillars: **non-interference in internal affairs** , a strict **commitment to strategic autonomy** , and avoiding involvement in each other's military alliances.
- In February 2026, during the visit of French President Emmanuel Macron to India for the **Artificial Intelligence Impact Summit 2026** and the launch of the **India-France Year of Innovation 2026** , the two countries elevated their bilateral ties to a **Special Global Strategic Partnership**.
- Long-term cooperation is guided by the **Horizon 2047 Roadmap** , celebrating a quarter-century of deep strategic trust.
- **Defence and Security Cooperation** France has firmly established itself as **India's second-largest arms supplier** , trailing only Russia.
 - Major procurement projects include a finalized contract for **26 Rafale-Marine fighter jets** , and Scorpene-class submarines.
 - A breakthrough in defense indigenization was achieved with the operationalization of the **H125 Helicopter Final Assembly Line in Karnataka** , a landmark joint venture between TATA and Airbus representing India's first private sector helicopter facility.
 - Institutionalized interoperability is maintained through three major annual tri-service exercises: **Shakti (Army)** , **Varuna (Navy)** , and **Garuda (Air Force)** .
- **Civil Nuclear and Space Collaboration:** Building upon the 2008 civil nuclear pact, both nations are partnering on cost-effective **Small Modular Reactors (SMRs)** and **Advanced Modular Reactors (AMRs)** under the 2025 Declaration of Intent.
 - France welcomed India's target of achieving **100 GW of nuclear power by 2047** and applauded the passage of the **SHANTI Act, 2025** , which permits private investment in the nuclear value chain to accelerate projects like the **Jaitapur Nuclear Power Plant**.
- **Joint Space Missions:** Deepening ISRO-CNES space ties, both nations are executing the **TRISHNA Mission** (Thermal Infrared Imaging Satellite for High-resolution Natural Resource Assessment) and scheduling the Third **India-France Strategic Space Dialogue for 2026**.
 - A Letter of Intent was signed between ISRO -Centre National d'Études Spatiales to strengthen cooperation in microgravity research and human space exploration.
- **Trade:** France stands as India's **3rd-largest trading partner within the European Union** (after the Netherlands and Germany), with bilateral trade reaching Euros 13.59 billion in 2025-26.
 - France is the 11th largest foreign investor in India; top equity inflows are concentrated heavily in the **services sector (17.65%)** , cement, and air transport.
- **Maritime Economy and Climate Action:** Guided by the **India-France Roadmap on Blue Economy and Ocean Governance (2022)** , both nations collaborate under the Indo-Pacific Oceans Initiative (IPOI) and the **Indian Ocean Rim Association (IORA)**.
 - Both nations are co-chairs of the **International Solar Alliance (ISA)** and the **Coalition for Disaster Resilient Infrastructure (CDRI)** for the 2024-26 cycle.
- **Cultural Infrastructure:** Ongoing development is underway for the **Yuge Yugeen Bharat Museum** via collaboration with France Muséums Développement, alongside plans for *Namaste France 2028* to mark 30 years of strategic partnership.

What are the Challenges in India-France Relations?

- **Stagnation of Legacy Mega-Projects:** The protracted stalling of the 9,900 MW Jaitapur Nuclear Power Project (JNPP) remains a glaring structural failure.
 - For over 15 years, the project has been paralyzed by techno-commercial disagreements regarding the **per-unit tariff of European Pressurized Reactors (EPRs)** and French apprehensions regarding India's stringent **Civil Liability for Nuclear Damage (CLND) Act, 2010**.
- **Geostrategic Asymmetries:** Despite a shared **Indo-Pacific vision**, both nations' primary threat matrices diverge.
 - France's immediate security preoccupations remain anchored in continental **Europe (the Ukraine conflict) and Francophone Africa**.
 - Conversely, India's existential focus is strictly on the Himalayas and the **Indian Ocean Region (IOR)**.
 - Furthermore, India's sustained strategic engagement with Russia occasionally **friction-tests European diplomatic sensitivities**.
- **Digital Governance and Regulatory Clashes:** As the partnership pivots heavily toward Artificial Intelligence and deep-tech, regulatory disconnect is emerging as a major non-tariff barrier.
 - India's developmental, open-architecture approach to Digital Public Infrastructure (e.g., DEPA) fundamentally misaligns with the European Union's ultra-restrictive, risk-averse frameworks, specifically the **GDPR** and the newly enforced **EU AI Act**.
- **Trade-Strategic Dissonance:** There is a stark asymmetry between the geopolitical depth of the partnership and its economic footprint.
 - Bilateral trade remains modest (USD 15.8 billion). Although political negotiations on the **India-EU Free Trade Agreement** have been completed, the agreement is yet to be implemented, delaying the full expansion of trade and investment flows.

What Measures can Strengthen India-France Relations?

- **Move Beyond Technology Transfer:** The Joint Advanced Technology Development Group should focus on co-developing advanced technologies such as jet engines, **Unmanned Combat Aerial Vehicles (UCAVs)**, and space-based defence systems with shared intellectual property and co-ownership.
 - Accelerate cooperation on SMRs under the 2025 Declaration of Intent rather than relying solely on the delayed Jaitapur project.
- **Trade and Economic Cooperation:** Use the Economic Security Dialogue to deepen cooperation in critical minerals, **semiconductors, green hydrogen, and resilient supply chains**.
- **Promote Balanced AI Governance:** Use the Joint AI Working Group to create a democratic and innovation-friendly approach to AI that balances safety with technological growth.
- **Deepen Trilateral Partnerships:** Strengthen India-France-Australia and India-France-UAE trilaterals through joint patrols, maritime exercises, and blue economy initiatives.
- **Enhance Maritime Security:** Expand logistics cooperation, intelligence sharing, and maritime domain awareness across the Indo-Pacific, leveraging French bases in the Indian Ocean region.

Conclusion

To bridge the gap between strategic intent and operational reality, the partnership must institutionalize dedicated bilateral bypass mechanisms—such as the newly formed Economic Security

Dialogue—to insulate critical supply chains and defense co-production from broader EU-level regulatory bottlenecks.

Drishti Mains Question:

"India-France relations have evolved from traditional strategic cooperation to a technology-driven partnership." Critically examine.

[Watch Video on YouTube:

▶ https://www.youtube.com/embed/9xIW_JijWhE

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Frequently Asked Questions (FAQs)

1. What is the Innovation Roadmap 2030 adopted by India and France?

It is a strategic framework aimed at enhancing technological sovereignty through cooperation in AI, digital infrastructure, semiconductors, critical technologies, innovation ecosystems, and startup collaboration.

2. What is the significance of the India-France Economic Security Dialogue?

The dialogue focuses on supply chain resilience, critical minerals, semiconductors, cybersecurity, energy security, and aims to expand bilateral trade and strategic economic cooperation.

3. Why is the Jaitapur Nuclear Power Project important in India-France relations?

JNPP is a flagship civil nuclear cooperation project involving European Pressurized Reactors (EPRs), but it faces delays due to tariff concerns and issues related to the Civil Liability for Nuclear Damage Act, 2010.

4. What is the Trusted AI Alliance between India and France?

It includes a Joint AI Working Group that promotes safe, ethical, transparent, and risk-based Artificial Intelligence governance while encouraging innovation and digital growth.

5. How does France contribute to India's defence modernization?

France is India's second-largest arms supplier and supports defence capability enhancement through Rafale aircraft, Scorpene submarines, defence co-production, aerospace manufacturing, and joint military exercises such as Shakti, Varuna, and Garuda.

Lt Gen Dhiraj Seth Appointed as Army Chief



Source: [PIB](#)

Lieutenant General Dhiraj Seth has been appointed the **next Chief of the Army Staff** and will assume office on 30th June 2026, succeeding General Upendra Dwivedi.

- **About:** He is a graduate of the **Higher Command Course** and the **National Defence College** , attended the **Command and Staff Course** in Paris, and also undertook defence acquisition training at the **United States Naval Postgraduate School**.
- **Career Distinction:** He will be the **third Army Chief** from the **2nd Lancers** after **General Maharaj Shri Rajendrasinhji Jadeja** and General Bipin Chandra Joshi.
 - He will be the **first Armoured Corps officer** to become Army Chief in nearly three decades after General Shankar Roy Chowdhury.
- **Army Command Roles:** He has commanded both **South Western Command** and **Southern Command**, a rare distinction of **leading two operational Army Commands** and providing strategic oversight across critical theatres for over two-and-a-half years.
- **Operational Legacy:** He headed Southern Command during **Operation Sindoor** , when formations were involved in securing Gujarat and Rajasthan sectors and countering Pakistani drones and aerial threats.
- **Honours:** He has served with the **United Nations Angola Verification Mission** and has been awarded the **Param Vishisht Seva Medal** , **Uttam Yudh Seva Medal** and **Ati Vishisht Seva Medal**.
- **Modernisation and Future Role:** He has held key staff and strategic appointments in the **Strategic Planning and Capability Development verticals** of Army Headquarters. As Army Chief, he is expected to **oversee technology absorption** , modern equipment induction, platform modernisation, force structuring and integration with the other services.



Read More: [Lt Gen NS Raja Subramani Appointed 3rd CDS](#)

Smart Village Initiative by CSIR



Source: TH

Recently, the **Council of Scientific and Industrial Research (CSIR)** is deploying **AI-enabled technologies** and infrastructure upgrades in Kusunpur village, Odisha, under the **‘Smart Village’ initiative** to mitigate **human-animal conflict**, enhance rural livelihoods, and integrate villages into India’s growth trajectory.

- **About:** An AI-enabled **wild animal detection and alert system** is being planned to **reduce human-animal conflict**, particularly **crocodile-related incidents**, by issuing alerts when crocodiles approach human settlements.
 - Kusunpur, one of **Odisha’s worst human-crocodile conflict zones with over 20 deaths in the past four years**, is being targeted under the initiative as a direct response to repeated wildlife-related fatalities.
- **‘Smart Village’ initiative:** It aims to **transform rural villages** by integrating **science, technology, and sustainable infrastructure**, improving quality of life, livelihoods, and safety.
- **Implementing Agency:** **Council of Scientific & Industrial Research (CSIR)** under its **lab-to-land approach**, deploying technologies developed in CSIR laboratories.
- **Key Features:**



- **Significance:**
 - Enhances **rural safety, disaster resilience, and human-animal conflict mitigation**.
 - Promotes **economic opportunities, entrepreneurship, and green jobs**.
 - Integrates villages into India’s **growth story** while demonstrating **applied science for social impact**.

Read More: [Nutrition Smart Village Initiative](#)

